Program 3: Benchmark Case 2

Description

Benchmark 2: This program performs the same task as Benchmark 1 but highly optimized using vector coprocessor instructions. It processes 4 students simultaneously by executing `VADD` vector instructions to parallelize the total computation. It applies vector clipping (`VCLIP`) and uses binary shifts to extract the final overall average in a fraction of the clock cycles.
Note: Each `IN` input will be a 32-bit value stored in a 32-bit register. It will go in order of all 20 IDs, all 20 M1s, and so on, in groups of 4.

Register Map:

- R1 = write pointer
- R2 = field counter
- R3 = ID
- R4 = M1
- R5 = M2
- R6 = M3
- R7 = M4
- R8 = Total
- R10 = Read IN
- R11 = Total DMEM offset
- R13 = Processed records
- R14 = DMEM offset pointer
- R15 = Current total
- R16 = Sum of totals
- R17 = Num records
- R18 = Average of totals
- R28 = total students (20)
- R29 = total count (100)
- R30 = Mark Cap (100)
- R1 = Write pointer
- R2 = field counter
- R3 = Vector 1 location
- R4 = Vector 2 location
- R5 = Vector 3 location
- R6 = Vector 4 location
- R7 = Vector 5 location
- R8 = Total Vector
- R9 = Current Vector
- R10 = Read Input
- R11 = Totals pointer

- R12 = Totals pointer offset
- R30 = Mark cap (100)
- R29 = total count (25)
- R28 = total students (20)
- R27 = inputs per type (5)
- R26 = Vector element 1 OR (F000)
- R25 = Vector element 2 OR (0F00)
- R24 = Vector element 3 OR (00F0)
- R23 = Vector element 4 OR (000F)
- R22 = Final total vectors (4)
- R21 = DIV shift numbers
- R20 = Result

Assembly Code	**Binary (Hex)**
`ADDI    R27, R27, 5`	277B0005
`ADDI    R28, R28, 20`	279C0014
`ADDI    R29, R29, 25 (100 values / 4 values per 32 bits = 25)`	27BD0019
`ADDI    R30, R30, 100`	27DE0064
Read_Loop:	
`   IN      R10`	300A0000
`   STORE   R10, 0(R1)`	202A0000
`   ADDI    R1, R1, 1`	24210001
`   ADDI    R2, R2, 1`	24420001
`   NOOP`	00000000
`   NOOP`	00000000
`   NOOP`	00000000
`   BLT     R2, R29, Read_Loop`	2C5DFFF8
`ADD     R11, R1, R0`	00205800
`ADD     R1, R0, R0`	00000800
`ADD     R2, R0, R0`	00001000
Proc_Loop:	
`   LOAD    R3, 5(R1)`	1C230005
`   LOAD    R4, 10(R1)`	1C24000A
`   LOAD    R5, 15(R1)`	1C25000F
`   LOAD    R6, 20(R1)`	1C260014
`   LOAD    R7, 25(R1)`	1C270019
`   VADD    R8, R3, R4`	40644000
`   VADD    R8, R8, R5`	41054000
`   VADD    R8, R8, R6`	41064000
`   VADD    R8, R8, R7`	41074000
`   VCLIP   R8, 100`	C1004780
`   ADD     R12, R11, R1`	01616000
`   STORE   R8, 0(R12)`	21880000
`   ADDI    R1, R1, 1`	24210001
`   NOOP`	00000000
`   NOOP`	00000000
`   NOOP`	00000000

```
Assembly Code                                                    Binary (Hex)
   BLT     R1, R27, Proc_Loop                                    2C3BFFEF


ADD     R1, R0, R0                                               00000800
ADD     R8, R0, R10                                              000A4000

Avg_Totals:
   ADD     R12, R11, R1                                          01616000
   LOAD    R8, 0(R12)                                            1D880000
   VADD    R9, R9, R8                                            41284800
   ADDI    R1, R1, 1                                             24210001
   NOOP                                                          00000000
   NOOP                                                          00000000
   NOOP                                                          00000000
   BLT     R1, R27, Avg_Totals                                   2C3BFFF8

ADDI    R26, R0, FF000000 (hex)                                  241A0000
ADDI    R25, R0, 00FF0000 (hex)                                  24190000
ADDI    R24, R0, 0000FF00 (hex)                                  2418FF00
ADDI    R23, R0, 000000FF (hex)                                  241700FF

OR      R26, R26, R9                                             0F49D000
OR      R25, R25, R9                                             0F29C800
OR      R24, R24, R9                                             0F09C000
OR      R23, R23, R9                                             0EE9B800

ADDI    R21, 01000000 (hex)                                      24150000
DIV     R26, R26, R21                                            1B55D000

ADD     R21, R0, R0                                              0000A800
ADDI    R21, 00010000 (hex)                                      24150000
DIV     R25, R25, R21                                            1B35C800

ADD     R21, R0, R0                                              0000A800
ADDI    R21, 00000100 (hex)                                      24150100
DIV     R24, R24, R21                                            1B15C000

ADD     R20, R23, R24                                            02F8A000
ADD     R20, R20, R25                                            0299A000
ADD     R20, R20, R26                                            029AA000
ADDI    R22, R0, 4                                               24160004

DIV     R20, R20, R22                                            1A96A000
OUT     R20                                                      36800000
```